

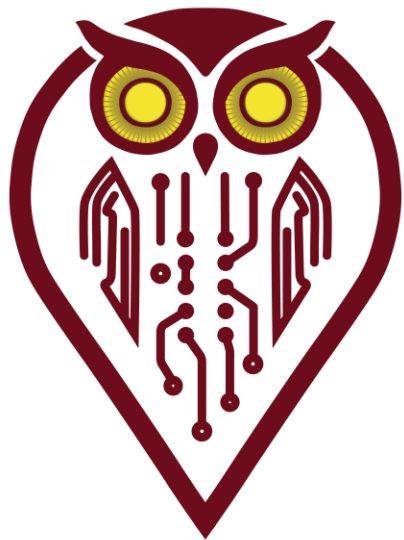
Advanced Machine Learning and Computer Vision

Advanced Machine Learning
25 September 2026
Prof. Fabio Galasso



SAPIENZA
UNIVERSITÀ DI ROMA

The lecturer and the research group



PINlab
PERCEPTION AND INTELLIGENCE

Lecture and Exercise

- Lectures:
 - ▶ Mondays 12:30-15:00 @Room B2 (Via Ariosto, 25) –lectures start at 12:30
 - ▶ Fridays 16:00-18:00 @Room B2 (Via Ariosto, 25) –lectures start at 16:15
- Office hours
 - ▶ Thursdays 16:00-18:00 @Room 43, 2° floor, build. G, via Regina Elena 295
- Website: <https://sites.google.com/di.uniroma1.it/amlcv-2026-2027>
- Mailing: https://groups.google.com/u/0/a/di.uniroma1.it/g/AMLCV_26_27
 - ▶ For slides, course material, assignments and news
- **Subscribe to it now with your institutional email**
 - ▶ Refer to the website home page (section “Subscribe”) if you do not have an institutional email address yet: your personal email address can be used temporarily upon proof of your acceptance to the degree program and proof of your identity

Exam

- Exam
 1. Theory: 50% (oral)
 2. Practical coding: 50%
 - Participation in a Kaggle challenge on a PINlab research topic
 - Follow-up research experience, presentation and brief report submission
- Assignments:
 - ▶ The practical coding **must be submitted in groups**
 - Groups must be **of size [2 – 4]**
 - **Find a team in class or post in the Google group**
 - ▶ To take part 1, you need a pass in part 2 of the exam
 - ▶ Practical coding marks remain valid for a calendar year
- Research experience
 - ▶ Groups will be free to choose specific research directions
 - ▶ Objectives and resources will be discussed in class

Assignment and final project

- Calendar
 - ▶ Challenge: 25 oct – 27 nov (~4 weeks)
 - ▶ Research experience: 27 nov - 18 dec (~3 weeks)
 - Your presentations: 11 and 18 Dec
 - Your report: by 21 Dec

Theory Exam

- Oral exam
 - ▶ 1 question chosen freely by the candidate
 - ▶ A few more questions, announced at the exam

Ethical Code of Conduct

- Plagiarism is an act of fraud
- Plagiarism is severely prohibited and, in any form, is regarded as a serious violation of the ethical code of conduct. Plagiarism includes the submission of source code and or report that bears strong resemblance to another team's or to resources online. After submission, every participation will be checked against plagiarism, including automatic detection tools
- Plagiarized submissions will be invalidated

Physical and Learning Disabilities

- Sapienza provides counseling and support
- You may reach out to:
sportellodisabili@uniroma1.it and counselingdsa@uniroma1.it
- Or directly to:
Prof. Tiziana Calamoneri
Coordinator for Disabilities and DSA for the I3S Faculty
<http://wwwusers.di.uniroma1.it/~calamo/>

Material

- Slides and coding scripts are distributed after lectures
- There is much material online (papers, blogs, twits, etc)
- Books (also: <https://sites.google.com/di.uniroma1.it/amlcv-2026-2027/resources>)
 - ▶ Machine Learning
 - Deisenroth, Faisal, Ong (2020). Mathematics for Machine Learning (<https://mml-book.github.io/book/mml-book.pdf>)
 - Christopher M. Bishop, 2006. Pattern Recognition and Machine Learning
 - ▶ Computer Vision
 - Antonio Torralba, Phillip Isola and William T. Freeman, 2024. Foundations of Computer Vision (<https://mitpress.mit.edu/9780262048972/foundations-of-computer-vision/>)
 - Richard Szeliski, 2010. Computer Vision: Algorithms and Applications (available at: <http://szeliski.org/Book>)
 - ▶ Robotics
 - Frank Dellaert, 2024, Robotics. (available at: <https://www.roboticsbook.org/>)

Material

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- Books (also: <https://sites.google.com/di.uniroma1.it/amlcv-2025-2026/resources>)
 - Deep learning
 - Aston Zhang, Zachary Lipton, Alexander J. Smola, Mu Li, 2023. Dive Into Deep Learning (available at: <https://d2l.ai/>)
 - Francois Fleuret, 2024. The Little Book of Deep Learning. (available at: <https://fleuret.org/dlc/>)
 - Michael M. Bronstein, Joan Bruna, Taco Cohen, Petar Veličković, 2024. Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (available at: <https://geometricdeeplearning.com/book/>)
 - Ian Goodfellow, Yoshua Bengio, Aaron Courville, 2017. Deep Learning (available at: <https://www.deeplearningbook.org/>)
 - Andrew Ng, 2019. Machine Learning Yearning (available at: <https://www.deeplearning.ai/machine-learning-yearning/>)

Coding References

- The practical coding will be in Python and Pytorch
- Books for Python and Pytorch
 - ▶ Aston Zhang, Zachary Lipton, Alexander J. Smola, Mu Li, 2023. Dive Into Deep Learning (available at <https://d2l.ai/> , select Notebooks/Pytorch)
 - ▶ Jake VanderPlas, 2016. Python Data Science Handbook: Tools and Techniques for Developers: Essential Tools for working with Data (Book and notebooks available at: <https://github.com/jakevdp/PythonDataScienceHandbook>)
- Online tutorials for Python: <https://docs.python.org/3/tutorial/>
- Online tutorials for Pytorch: <https://pytorch.org/tutorials/>
 - ▶ <https://sebastianraschka.com/teaching/pytorch-1h/>

Setup and computing

- For the practical coding a GPU is recommended
 - Consider Google colab: <https://colab.research.google.com>
 - and Kaggle: <https://www.kaggle.com/>

Pre-requisites and recommendations

- Proficiency in Python, some experience with Pytorch
 - The class assignment will be in Python and Pytorch
- Calculus, Linear Algebra, Probability
 - Taking derivatives, understanding matrix vector operations and notation
- Machine Learning
 - Regression, binary and multi-class classification
 - Cost functions, derivatives and optimization with gradient descent
- Basics of Deep Learning
 - Basics of fully-connected, convolutional and recurrent networks
 - Basics of backprop, learning and optimization for neural networks
 - Basics of attention and Transformer Networks

Syllabus

- Advanced sequence modelling for long-term and structured data with ConvNets, RNN/LSTM, Transformers, Graph Convolutional Networks and State-Space Models, and applications to human motion forecasting, earthquake forecasting and coding
- Semantic segmentation, object detection, instance segmentation, human pose estimation and tracking
- Self-supervised, meta-, continual and deep reinforcement learning
- Generative AI including VAEs, GANs, diffusion and flow matching models, and applications to anomaly detection and human motion and video synthesis
- Hyperbolic Neural Networks, representing hierarchies and learning with uncertainty, Multi-modal Language Models
- World Action Models and ML for robotics, safe AI, machine unlearning and model editing

Thank you



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